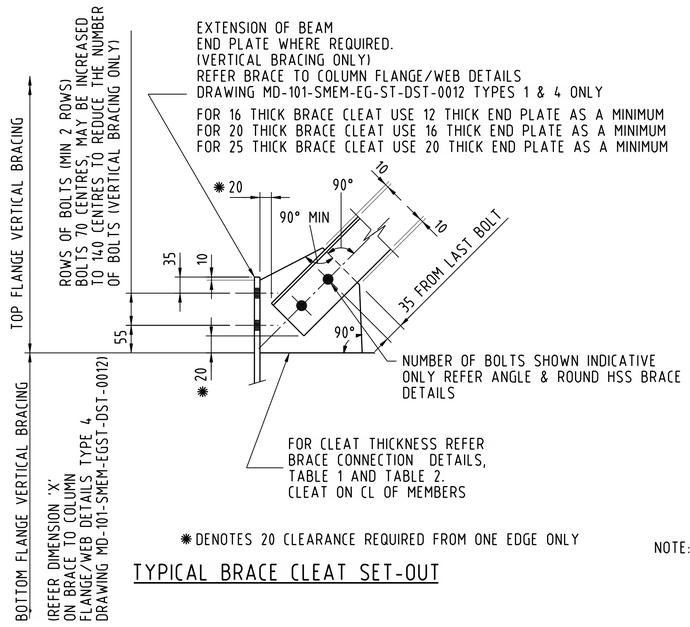
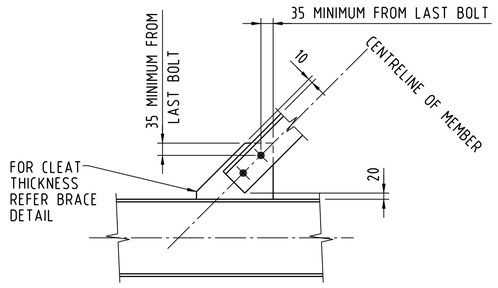


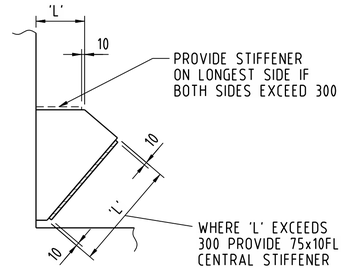
✸ DENOTES 20 CLEARANCE REQUIRED FROM ONE EDGE ONLY

TYPICAL BRACE CLEAT SET-OUT



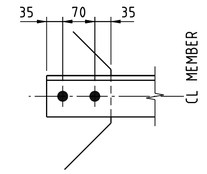
ALTERNATIVE BRACING CLEAT DETAIL

FOR ANGLE BRACES L4 AND SMALLER

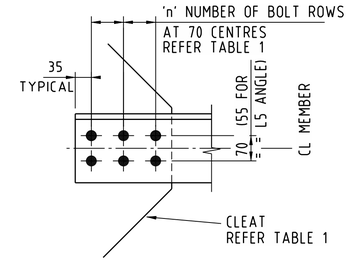


TYPICAL VERTICAL BRACING CLEAT STIFFENER DETAIL

TABLE 1		
BRACING MEMBER SIZE	CLEAT THICKNESS	NUMBER OF BOLT ROWS 'n'
L2.5x2.5x0.1875 TO L3.5x3.5x0.375	10	2
L2.5x2x0.1875 TO L3x2x0.3125	10	2
L4x4x6 TO L6xL6x0.5	12	2
L4x3x0.25 TO L6x4x0.5	12	2
L6x6x0.625 TO L8x8x1	16	3



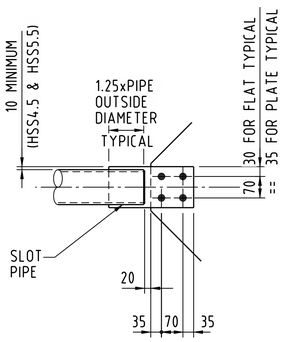
L4 ANGLE OR SMALLER



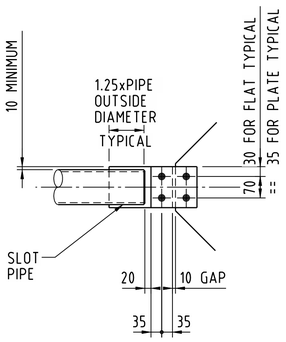
L5 ANGLE OR LARGER

BRACING CLEAT DETAIL

NOTE: 1. FOR BRACE END DETAILS REFER TO ANGLE AND ROUND HSS BRACE DETAILS
2. BRACING CLEATS SHALL BE CENTRAL ON CL BEAM/COLUMN. (VERTICAL BRACING ONLY)

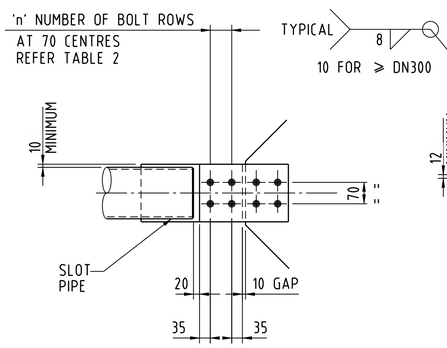


NON PREFERRED



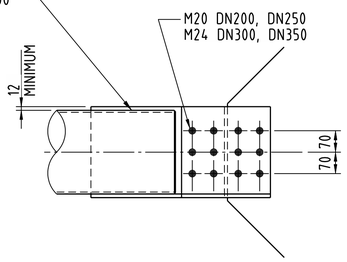
PREFERRED

DN80-STANDARD AND DN100-STANDARD



DN125-STANDARD, DN150-STANDARD

FOR DETAILS NOT SHOWN
REFER DN80 DETAIL ADJACENT



DN200-STANDARD, DN250-STANDARD,
DN300-STANDARD
DN300-EXTRA STRONG, DN350-EXTRA STRONG

FOR DETAILS NOT SHOWN REFER
DN80 AND DN125 DETAILS ADJACENT

PIPE BRACE CONNECTION DETAIL

NOTE:
WHERE UNEQUAL ROUND HSS SIZES CONNECT TO THE SAME CLEAT, CLEAT AND TONGUE PLATE THICKNESS SHALL BE DETERMINED BY THE LARGER CLEAT THICKNESS

TABLE 2			
BRACING MEMBER SIZE	CLEAT/TONGUE PL THICKNESS	COVER PLATE THICKNESS	NUMBER OF BOLT ROWS 'n'
DN80-STANDARD	12	10	1
DN100-STANDARD	12	10	1
DN125-STANDARD	20	10	2
DN150-STANDARD	20	10	2
DN200-STANDARD	25	12	2
DN200-EXTRA STRONG	25	12	2
DN250-STANDARD	32	16	2
DN250-EXTRA STRONG	32	16	2
DN300-STANDARD	36	20	3
DN300-EXTRA STRONG	36	20	3
DN350-EXTRA STRONG	36	20	3

LEGEND ANGLE AND PIPE BRACES

DENOTED ON DESIGN DRAWINGS FOR NON-STANDARD CONNECTIONS

n/T/p
- DENOTES BOLTING PROCEDURE (OTHER THAN S)
- T FOLLOWED BY A NUMBER DENOTES CLEAT/TONGUE PL THICKNESS OTHER THAN STANDARD
- DENOTES NUMBER OF BOLT ROWS OTHER THAN STANDARD

NOTES:

1. FOR STEELWORK GENERAL NOTES REFER DRAWING MD-101-SMEM-EG-ST-DST-0009.
2. STRUCTURAL CAPACITIES OF THE CONNECTIONS SHOWN ON THIS DRAWING SHALL BE CHECKED BY THE CONTRACTOR TO DETERMINE THEIR SUITABILITY FOR THE INTENDED USE.
3. FOR TABLES 1 & 2, REFER TO MD-101-SMEM-EG-ST-SPC-0001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	MD

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REV C01 PREPARED FOR MA'ADEN BY HATCH ASSOCIATES H332701-0000-35-041-0013_C

MA'ADEN معادن
Saudi Arabian Mining Company

DRAWN	J BADAL	04.03.11
CHECKED	J EDWARDS	18.04.12
ENGINEER	D KIRKWOOD	18.04.12
APPROVED	R COVELLA	18.04.12
MA'ADEN		
SCALE NTS		A1

MA'ADEN ENGINEERING MANUAL
ALL AREAS
STRUCTURAL STANDARDS
BRACE CONNECTIONS SHEET 2

DRAWING NUMBER MD-101-SMEM-EG-ST-DST-0013 REV C01

MA'ADEN PHOSPHATE PROJECT-A1 BORDER-REV A02

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